```
!pip install -q groq
```

```
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 143.7/143.7 kB 5.0 MB/s eta 0:00:00
```

```
from google.colab import userdata
import os

api_key = userdata.get("Secret_Key")

if not api_key:
    raise ValueError("Secret_Key was not found in Colab Secrets")

os.environ["GROQ_API_KEY"] = api_key

print("API key loaded successfully")
```

```
API key loaded successfully
```

```
from groq import Groq
import os

client = Groq(
    api_key=os.environ["GROQ_API_KEY"]
)

MODEL = "llama-3.3-70b-versatile"


def run_agent(agent_name, system_prompt, user_input):

    response = client.chat.completions.create(
        model=MODEL,
        messages=[
            {
                "role": "system",
                "content": system_prompt
            },
            {
                "role": "user",
                "content": user_input
            }
        ],
        temperature=0.3
    )

    result = response.choices[0].message.content

    print("\n" + "=" * 70)
    print(agent_name)
    print("=" * 70)
    print(result)

    return result
```

```
research_prompt = """
You are the Research Agent in a multi-agent collaboration system.

Your responsibility is to break the user's task into important information
requirements and provide accurate background information.

Do not write the final answer.
Focus only on research, facts, important concepts, and useful points.
"""


analysis_prompt = """
You are the Analysis Agent.

You receive research from another agent.

Your responsibility is to analyze the research, identify the most important
points, remove unnecessary information, identify relationships between ideas,
and create a logical structure for the final response.

Do not produce the final polished answer.
```

```
"""


writer_prompt = """
You are the Writer Agent.

You receive the original task, research, and analysis from other agents.

Your responsibility is to create a clear, well-structured answer to the
original task.

Use simple language and organize the response logically.
"""


reviewer_prompt = """
You are the Reviewer Agent.

You receive a draft answer created by another agent.

Check the answer for correctness, clarity, missing information,
contradictions, and unnecessary content.

Then produce an improved final answer.

Your output must be the final answer only.
"""
```

```
def multi_agent_system(task):

    print("\n" + "#" * 70)
    print("MULTI-AGENT COLLABORATION SYSTEM")
    print("#" * 70)

    print("\nUSER TASK:")
    print(task)

    # Agent 1: Research
    research = run_agent(
        "AGENT 1 - RESEARCH AGENT",
        research_prompt,
        task
    )

    # Agent 2: Analysis
    analysis_input = f"""
Original Task:
{task}

Research Agent Output:
{research}
"""

    analysis = run_agent(
        "AGENT 2 - ANALYSIS AGENT",
        analysis_prompt,
        analysis_input
    )

    # Agent 3: Writer
    writer_input = f"""
Original Task:
{task}

Research:
{research}

Analysis:
{analysis}
"""

    draft = run_agent(
        "AGENT 3 - WRITER AGENT",
        writer_prompt,
        writer_input
    )

    # Agent 4: Reviewer
    reviewer_input = f"""
Original Task:
{task}
```

```
Research:
{research}

Analysis:
{analysis}

Draft Answer:
{draft}
"""

    final_answer = run_agent(
        "AGENT 4 - REVIEWER AGENT",
        reviewer_prompt,
        reviewer_input
    )

    return final_answer
```

```
task = """
Explain how Artificial Intelligence is used in Cybersecurity.
Discuss major applications, advantages, limitations, and future scope.
"""

final_answer = multi_agent_system(task)
```

```
######################################################################
MULTI-AGENT COLLABORATION SYSTEM
######################################################################

USER TASK:

Explain how Artificial Intelligence is used in Cybersecurity.
Discuss major applications, advantages, limitations, and future scope.


======================================================================
AGENT 1 - RESEARCH AGENT
======================================================================
**Introduction to AI in Cybersecurity**

Artificial Intelligence (AI) is being increasingly used in cybersecurity to enhance the detection, prevention, and respons

**Major Applications of AI in Cybersecurity**

1. **Intrusion Detection Systems (IDS)**: AI-powered IDS can detect and alert on potential threats in real-time, reducing
2. **Predictive Analytics**: AI algorithms can analyze historical data to predict and prevent future cyber attacks.
3. **Threat Intelligence**: AI-powered systems can analyze vast amounts of threat data to identify patterns and predict fu
4. **Incident Response**: AI can help automate incident response processes, reducing the time and cost associated with res
5. **Security Information and Event Management (SIEM) Systems**: AI-powered SIEM systems can analyze log data to identify
6. **Endpoint Security**: AI-powered endpoint security solutions can detect and prevent malware and other threats on endpo
7. **Network Security**: AI-powered network security solutions can detect and prevent threats in real-time, reducing the r

**Advantages of AI in Cybersecurity**

1. **Improved Detection**: AI-powered systems can detect threats that may evade traditional security systems.
2. **Increased Efficiency**: AI can automate many security tasks, freeing up security teams to focus on more strategic tas
3. **Enhanced Incident Response**: AI can help security teams respond to incidents more quickly and effectively.
4. **Predictive Capabilities**: AI can predict and prevent future cyber attacks, reducing the risk of security breaches.
5. **Scalability**: AI-powered systems can analyze vast amounts of data, making them ideal for large-scale security deploy

**Limitations of AI in Cybersecurity**

1. **Data Quality**: AI-powered systems require high-quality data to function effectively.
2. **Bias and Error**: AI algorithms can be biased or flawed, leading to incorrect conclusions and decisions.
3. **Lack of Transparency**: AI-powered systems can be difficult to understand and interpret, making it challenging to ide
4. **Dependence on Human Expertise**: AI-powered systems require human expertise to train, configure, and maintain them.
5. **Evasion Techniques**: Sophisticated attackers can use evasion techniques to evade AI-powered security systems.

**Future Scope of AI in Cybersecurity**

1. **Increased Adoption**: AI is expected to become more widespread in cybersecurity, with more organizations adopting AI-
2. **Advancements in Machine Learning**: Advancements in machine learning algorithms are expected to improve the accuracy
3. **Integration with Other Technologies**: AI is expected to be integrated with other technologies, such as Internet of T
4. **Development of New Threats**: The increasing use of AI in cybersecurity is expected to lead to the development of new
5. **Growing Importance of Human-AI Collaboration**: The future of cybersecurity is expected to involve increased collabor

**Key Concepts and Technologies**

1. **Machine Learning**: A subset of AI that involves training algorithms on data to make predictions or decisions.
2. **Deep Learning**: A subset of machine learning that involves the use of neural networks to analyze data.
3. **Natural Language Processing (NLP)**: A subset of AI that involves the analysis and interpretation of human language.
```

```
Start coding or generate with AI.
```